

Battery door (LiPo 503035)



#	Part no.	Description	Material	Qty	Process / tool
1	HS-101	Front housing	ABS white	1	Inj. mould - Tool A (cav 1)
2	HS-102	Rear housing	ABS white	1	Inj. mould - Tool A (cav 2)
3	HS-103	Battery door	ABS white	1	Inj. mould - Tool A (cav 3)
4	HS-104	Power button cap	ABS black	1	Inj. mould - Tool B (cav 1)
5	HS-105	Optical block (light-tight)	ABS black	1	Inj. mould - Tool B (cav 2)
5a	HS-106	Strip tray (slide-in, detent)	ABS black	1	Inj. mould - Tool B (cav 3)
6	HS-201	Display lens 33.8x40.8x0.8	PMMA, HC one side	1	Laser cut + 3M 9471LE
7	HS-202	Graphic overlay 54.6x114.6	PC 0.175 + 7952MP	1	Screen print + die cut
8	EJOT PT K25x16	Housing screw	Steel Zn	4	Torque 0.35 Nm
9	EJOT PT K25x6	PCB + optical block screw	Steel Zn	4	Torque 0.25 Nm
10	3M SJ5302	Bumpon foot dia 7.9	PU	4	-
11	(PCB)	Main PCB 76.75x49.94x1.6	FR4	1	thk 1.6 +/-0.10 (spec!)
12	(bought)	1.8in TFT ST7735, LiPo 503035, 6x6x5 tact	-	1 ea	per electronics BOM

1. Datum A = parting plane (P/L), z = 14.2 from the rear skin. B = centre line X. C = centre line Y.
2. All plastics: ABS, should shrink 0.5 % built into the tool STEP data (see HS-300 sheets).
3. Parting plane runs through the USB-C centre and the strip-tray mouth: No slides / lifters in any tool.
- 3a. SINGLE BUTTON. Strip goes in a slide-in tray (HS-106): load strip outside, push tray home (hard stop + detent click); the tray end stop and the read aperture are on the same black part (HS-105).
4. Housing closure: rim lap joint + 4 x PT K25x16 from the rear. Top two screws pass through the PCB top holes (PCB NOT clamped: 0.30 hold-down gap). Bottom PCB holes: 2 x PT K25x6.
5. White ABS is translucent: the optical path is enclosed in the BLACK optical block (HS-105).
6. Tolerances: general per sheet 6; critical fits and 15 stack-ups on sheet 6.
7. Colour variants (teal / grey / purple ...) come from the printed overlay only - one set of tools.
8. OPEN ITEMS: confirm PCB component heights (<= 3.0 below, TFT FPC route), USB-C overhang 3.0 mm past PCB edge, tact switch at (0, -8), HB STRIP SIZE (6.0 x 30 x 0.5 assumed), LiPo 503035 wiring.

LEVRAM LIFESCIENCES

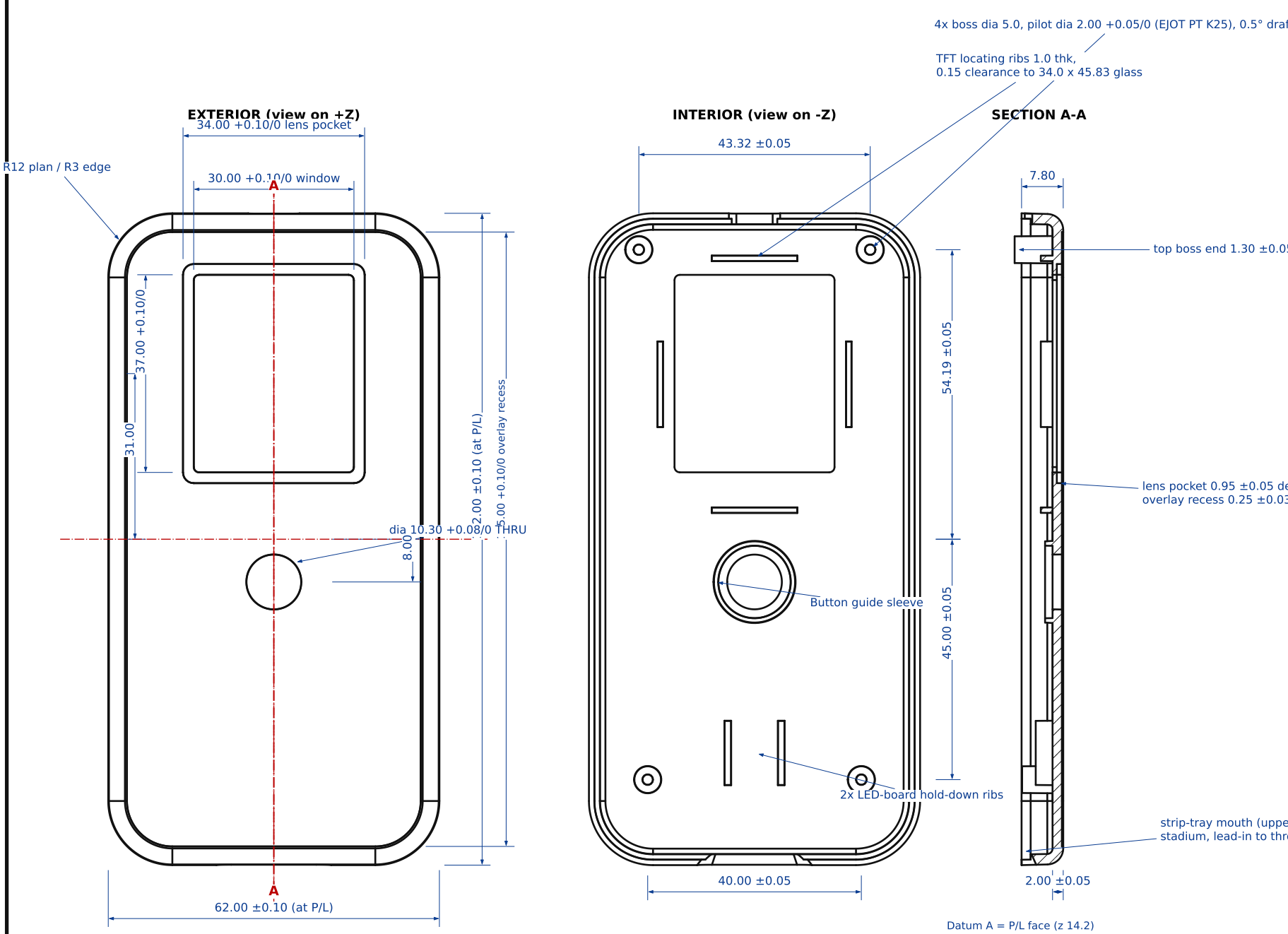
GENERAL ARRANGEMENT, STACK-UP & BILL OF MATERIALS

HemoSure Hb meter casing

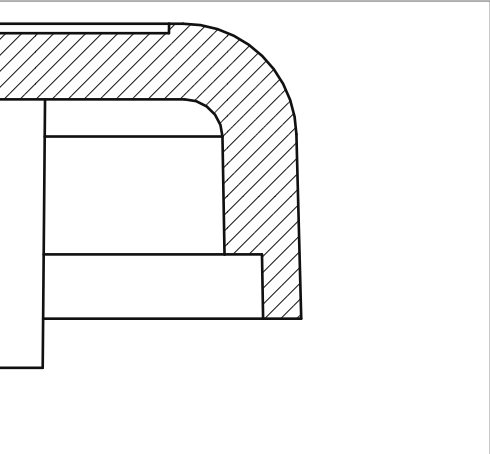
Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART HemoSure casing assembly	DWG NO. HS-100	REV A
MATERIAL see BOM	SCALE 1:1 / NTS	SHEET 1 / 11
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)
MASS 42 g (plastics)	DATE 2026-09-28	STATUS DRAFT - DFM review

A
B
C
D
E
F



DETAIL B - LAP STEP (5:1, section y = 0)



CRITICAL DIMENSIONS (HS-101)

Feature	Nominal	Tol.	Steel-safe
Outer profile at P/L	62.00 x 122.00	+0.10 / -0.10	-
Display window (at pocket floor)	30.00 x 37.00	+0.10 / 0	cut small
Lens pocket	34.00 x 41.00	+0.10 / 0	cut small
Lens pocket depth	0.95	+0.05 / -0.05	cut shallow
Overlay recess	55.00 x 115.00	+0.10 / 0	cut small
Overlay recess depth	0.25	+0.03 / -0.03	cut shallow
Button hole	dia 10.30	+0.08 / 0	cut small
Screw pilot (PT K25)	dia 2.00	+0.05 / 0	pin small
Top boss end below P/L	1.30	+0.05 / -0.05	leave long
Bottom boss end above P/L	0.10	+0.05 / -0.05	leave long
Lap step width x depth	1.00 x 1.70	+0.05 / 0	cut narrow
Boss pattern (both halves)	43.32 x 70.13	+0.05 / -0.05	-

NOTES

- Material ABS natural white (colour per overlay). Mould shrink 0.5 % (tool data). Wall 2.00 nominal.
- Draft: outer walls 1.5° (texture VDI 24), inner walls 1.0°, bosses/ribs 0.5°/side - modelled, do not add.
- Ribs 1.0-1.2 thk (<= 60 % wall) - no sink on the cosmetic face. Boss OD 5.0 / pilot 2.0 (wall 1.5).
- P/L flatness 0.10. Outer profile at P/L to match HS-102 within 0.10 per side (step).
- No flash on P/L, lap step and window; no sink / flow marks / weld line on the overlay area.
- Gate: 1 x sub (tunnel) gate dia 1.2 into inner +X wall at y = 0 (hidden). Ejector marks core side only, flush +0/-0.05.
- General tolerances: sheet 6 table (ISO 20457-style). Fits marked here override.

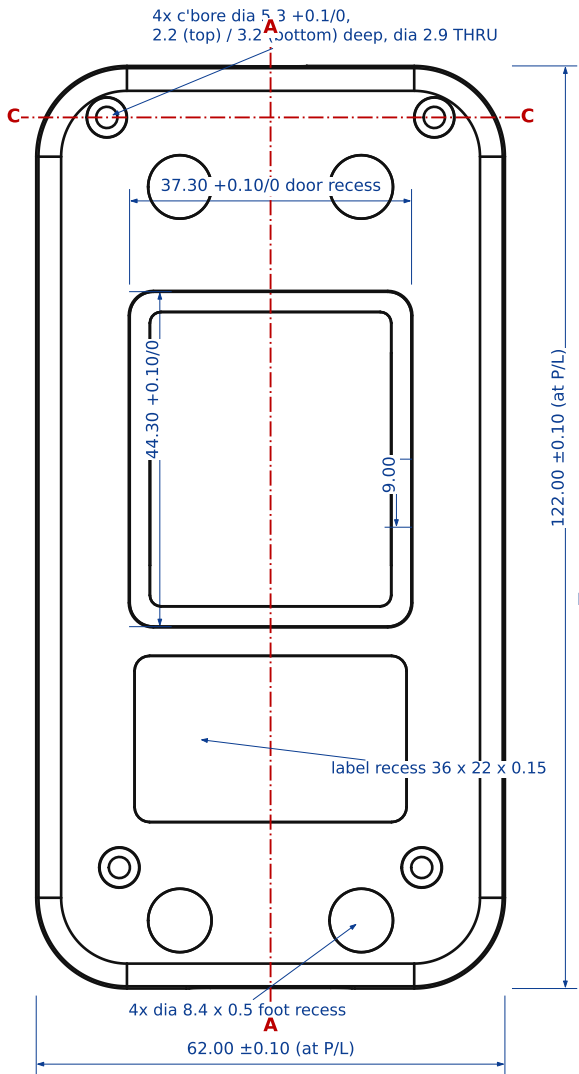
LEVRAM LIFESCIENCES
FRONT HOUSING

HemoSure Hb meter casing
Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

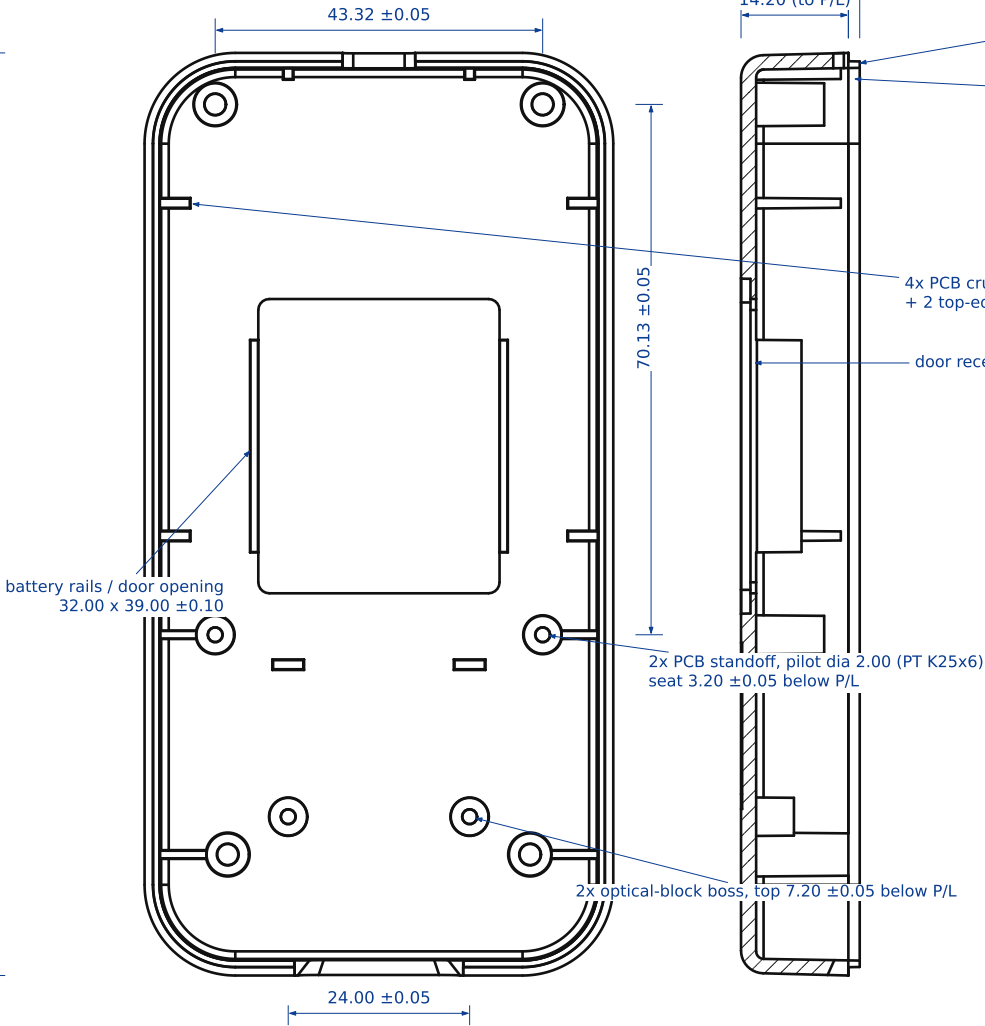
PART Front housing	DWG NO. HS-101	REV A
MATERIAL ABS natural white (LG HI-121H / PA-757 eq.)	SCALE 1:1	SHEET 2 / 11
FINISH Out VDI 3400 Ref 24 / In SPI B-2	UNITS mm	PROJ. 1st angle (ISO E)
MASS 15.0 g	DATE 2026-09-28	STATUS DRAFT - DFM review

A
B
C
D
E
F

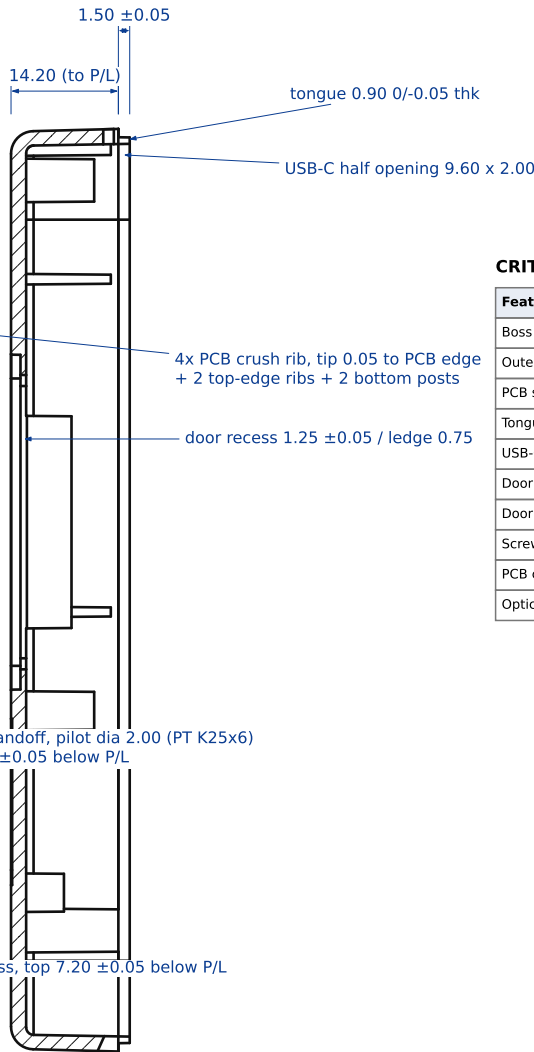
EXTERIOR (view on -Z)



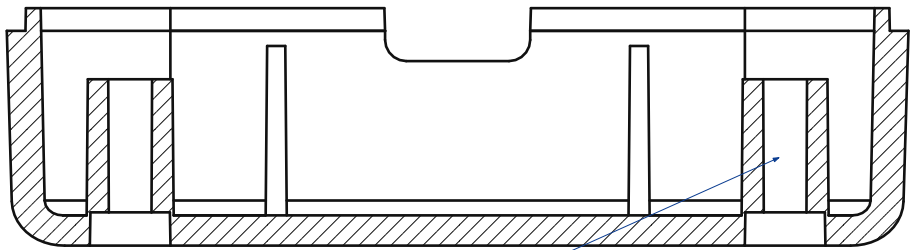
INTERIOR (view on +Z)



SECTION A-A



SECTION C-C (y = 54.19, through top screw path) 2:1



CRITICAL DIMENSIONS (HS-102)

Feature	Nominal	Tol.	Steel-safe
Boss pattern (both halves)	43.32 x 70.13	+0.05 / -0.05	-
Outer profile at P/L	62.00 x 122.00	+0.10 / -0.10	-
PCB seat below P/L	3.20	+0.05 / -0.05	leave long
Tongue thickness x height	0.90 x 1.50	0 / +0.05 / -0.05 / -0.05	cut thick
USB-C opening (split on P/L)	9.60 x 4.00	+0.10 / 0	cut small
Door recess	37.30 x 44.30	+0.10 / 0	cut small
Door recess depth	1.25	+0.05 / -0.05	cut shallow
Screw clearance / c'bore	dia 2.9 / dia 5.3	+0.10 / 0	-
PCB crush-rib tip to PCB edge	0.05	+0.05 / -0.05	cut tight
Optical-block boss top below P/L	7.20	+0.05 / -0.05	leave long

NOTES

- Material ABS natural white, shrink 0.5 %. Wall 2.00; floor 2.00; door ledge 0.75 (reinforced by rails).
- Draft as HS-101 (modelled). Tongue 1° both faces. Recesses 1°. Pins/holes 0.25-1°.
- PCB located by crush ribs (+/-0.05) and 2 PT K25x6 at the bottom holes; top holes carry the housing screws.
- USB-C and strip-tray mouth split on the P/L: half in each housing, no side action.
- Gate: 1 x sub gate dia 1.2 into inner +X wall at y = +20 (hidden). Ejection: sleeves on all bosses.
- No sink opposite bosses / rails on the rear skin (bosses cored, gussets 1.0 thk).
- Label recess and foot recesses: keep texture-free (SPI B-2) for adhesion.

LEVRAM LIFESCIENCES

HemoSure Hb meter casing

REAR HOUSING

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART Rear housing	DWG NO. HS-102	REV A
MATERIAL ABS natural white (LG HI-121H / PA-757 eq.)	SCALE 1:1	SHEET 3 / 11
FINISH Out VDI 3400 Ref 24 / In SPI B-2	UNITS mm	PROJ. 1st angle (ISO E)
MASS 22.3 g	DATE 2026-09-28	STATUS DRAFT - DFM review

1 2 3 4 5 6 7 8

A



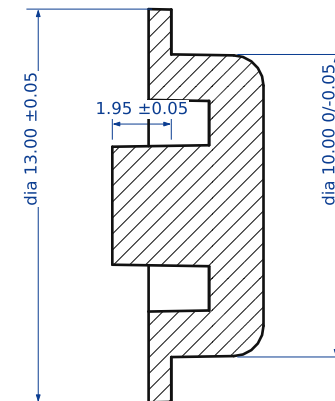
C



HS-103 SECTION (x = 0)



HS-104 BUTTON - section 4:1



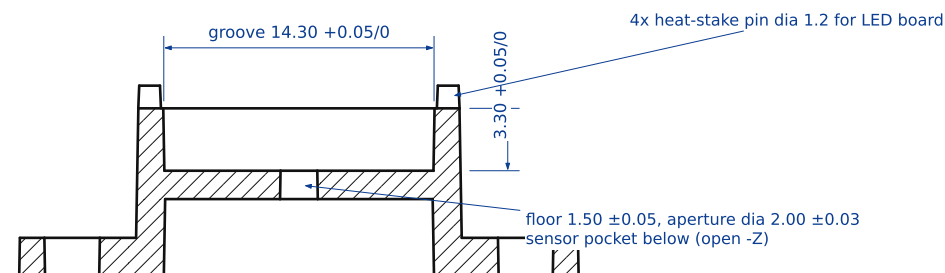
NOTES

HS-103 door: ABS white, same resin lot as HS-102.
 Flush 0/-0.15 to rear skin; gap 0.15-0.25/side.
 HS-104 button: ABS black (or TPU 90A if softer
 feel wanted - same tool). Pad-print power icon.
 HS-105 optical black: ABS BLACK, opaque at 2 mm
 (transmission < 0.1 % @ 400-700 nm - verify).
 Groove Ra <= 0.8 (SPI B-1) to avoid scratching
 the strip tray. Parting at groove end.
 Add 0.5 x 45° lead-in at groove entry (T1).
 All: draft 1° modelled on every wall.

Cap rests on the tact actuator; 0.45 play to the housing face (stack 3).

Mould windows under hooks/bump let the cavity form the hook undersides (no lifters).

HS-105 SECTION D-D (y = -40, optical axis) 2.5:1

F

A

B

C

D

E

F



A

B

C

D

E

F



A

B

C

D

E

F

A

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A

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F

1		2
GENERAL TOLERANCES - ABS (unless stated)		
Size range (mm)	Tool-bound (single half)	Across P/L / not tool-bound
0 - 6	+/-0.10	+/-0.15
> 6 - 30	+/-0.15	+/-0.20
> 30 - 120	+/-0.25	+/-0.35
> 120 - 315	+/-0.40	+/-0.55

BASIS

Basis: ISO 20457 tolerance groups for ABS (shrink 0.4-0.7 %), normal production.
 Angles $\pm 0.5^\circ$. Radii ± 0.2 . Flatness P/L 0.10. Position of holes ± 0.05 where on the PCB / screw pattern, else general. Critical dims are T1-tuned ('steel-safe').
 Steel-safe = cut the tool so that metal can only be REMOVED to reach nominal:
 holes/pockets cut small, bosses/pins left long, recesses cut shallow.

CRITICAL / FUNCTIONAL DIMENSIONS

Part	Feature	Nominal	+	-
Front	Outer profile at P/L	62.00 x 122.00	+0.10	-0.10
Front	Display window (at pocket floor)	30.00 x 37.00	+0.10	0
Front	Lens pocket	34.00 x 41.00	+0.10	0
Front	Lens pocket depth	0.95	+0.05	-0.05
Front	Overlay recess	55.00 x 115.00	+0.10	0
Front	Overlay recess depth	0.25	+0.03	-0.03
Front	Button hole	dia 10.30	+0.08	0
Front	Screw pilot (PT K25)	dia 2.00	+0.05	0
Front	Top boss end below P/L	1.30	+0.05	-0.05
Front	Bottom boss end above P/L	0.10	+0.05	-0.05
Front	Lap step width x depth	1.00 x 1.70	+0.05	0
Front/Rear	Boss pattern (both halves)	43.32 x 70.13	+0.05	-0.05
Rear	Outer profile at P/L	62.00 x 122.00	+0.10	-0.10
Rear	PCB seat below P/L	3.20	+0.05	-0.05
Rear	Tongue thickness x height	0.90 x 1.50	0 / +0.05	-0.05 / -0.05
Rear	USB-C opening (split on P/L)	9.60 x 4.00	+0.10	0
Rear	Door recess	37.30 x 44.30	+0.10	0
Rear	Door recess depth	1.25	+0.05	-0.05
Rear	Screw clearance / c'bore	dia 2.9 / dia 5.3	+0.10	0
Rear	PCB crush-rib tip to PCB edge	0.05	+0.05	-0.05
Rear	Optical-block boss top below P/L	7.20	+0.05	-0.05
Door	Plate outline x thickness	37.00 x 44.00 x 1.20	0	-0.10 / -0.05
Door	Latch arm t x bump reach	0.80 x 0.80	+0.03 / +0.05	-0.03 / -0.05
Button	Cap dia / flange dia	10.00 / 13.00	0 / +0.05	-0.05 / -0.05
Button	Stem tip to flange top	1.95	+0.05	-0.05
Optical	Tray groove width x depth	14.30 x 3.30	+0.05	0
Optical	End stop to aperture centre	15.00	+0.05	-0.05
Tray	Body width x thickness	14.10 x 3.10	0	-0.05 / -0.05
Tray	Strip pocket W x L x D	6.10 x 30.10 x 0.75	+0.05 / +0.1 / +0.080	-0.05 / -0.03
Tray	Retention rib depth / pocket recess	0.13 / 0.25	+0.02 / +0.03	-0.02 / -0.03
Tray	Front face to pocket wall	1.00	+0.05	-0.05
Back	Detect bump reach / notch depth	45 / 0.40	+/-0.05	+/-0.05
Optical	Floor under groove / aperture holes dia 3.20 ±0.05; Hole pattern ±0.05; USB-C overhang 3.0 ±0.1.	1.50 / dia 2.00	+/-0.05	+/-0.03
Port switch 6x6, height 5.00±0.10, travel 0.25, 1.6 N.		24.00	+0.05	-0.05

USB-C receptacle 8.94 x 3.26 shell, top-mount, centre 1.63 above PCB.
TFT 1.8in ST7735: glass 34.00 x 45.83 ±0.10, active area offset
2.0 towards the FPC-free end (confirm module drawing).
Hb strip 6.00 ±0.05 x 30.0 x 0.50 ±0.03, read window 13.95 ±0.2
from the front edge (ASSUMED - confirm strip drawing).

LATCH STRAIN (HS-103): $e = 1.5 t d / L^2$, $t \geq 0.8$, $L \geq 6.3$:
nominal $d \geq 0.5 \rightarrow 1.5\%$; max $d \geq 0.75 \rightarrow 2.3\%$ (ABS limit $\sim 2.5\%$).
SCREW BOSSES: PT K25 in ABS, pilot 2.0, engagement ≥ 5.0 (2d).

STACK-UP ANALYSIS (gaps in mm; + = clearance)

#	Stack	Nom.	Worst case	RSS	Accept	WC	RSS
1	Housing closure: front vs rear bottom screw bosses	+0.100	+0.00 .. +0.20	+0.03 .. +0.17	+0.00 .. +0.25	OK	OK
2	PCB hold-down gap (front top boss to PCB top)	+0.300	+0.04 .. +0.56	+0.13 .. +0.47	+0.00 .. +0.60	OK	OK
3	Button: flange-to-face play with cap resting on the switch	+0.450	+0.10 .. +0.80	+0.28 .. +0.62	+0.05 .. +0.90	OK	OK
4	Button protrusion above the front face at rest	+0.600	+0.15 .. +1.05	+0.20 .. +1.00	+0.15 .. +1.10	OK	OK
5	Lens below overlay floor (flush)	+0.100	-0.01 .. +0.21	+0.03 .. +0.17	-0.02 .. +0.25	OK	OK
6	Overlay flush in its recess	+0.025	-0.03 .. +0.08	-0.01 .. +0.06	-0.03 .. +0.10	OK	OK
7	Battery door flush with the rear skin	+0.050	-0.05 .. +0.15	-0.02 .. +0.12	-0.05 .. +0.20	OK	OK
8	Battery door gap per side	+0.150	+0.05 .. +0.25	+0.08 .. +0.22	+0.05 .. +0.30	OK	OK
9	Door latch engagement over the ledge	+0.500	+0.20 .. +0.80	+0.33 .. +0.67	+0.20 .. +0.80	OK	OK
10	Lap joint radial clearance (tongue to front step)	+0.100	-0.01 .. +0.21	+0.03 .. +0.17	-0.02 .. +0.25	OK	OK
11	Lap joint: tongue height vs step depth	+0.200	+0.10 .. +0.30	+0.13 .. +0.27	+0.05 .. +0.35	OK	OK
12	USB-C: vertical clearance in the opening (worst side)	+0.370	+0.09 .. +0.65	+0.16 .. +0.58	+0.00 ..	OK	OK
13	USB-C: lateral clearance in the opening (worst side)	+0.330	+0.00 .. +0.66	+0.14 .. +0.52	+0.00 ..	OK	OK
14	PCB hole vs M2.5 housing screw (top holes)	+0.350	+0.00 .. +0.70	+0.16 .. +0.54	+0.00 ..	OK	OK
15	Strip tray lateral clearance in the groove (per side)	+0.100	+0.05 .. +0.15	+0.06 .. +0.14	+0.03 .. +0.20	OK	OK
16	Strip tray vertical clearance under the LED board	+0.200	+0.10 .. +0.30	+0.13 .. +0.27	+0.05 .. +0.35	OK	OK
17	Strip lateral clearance in the tray pocket (per side)	+0.050	-0.00 .. +0.10	+0.01 .. +0.09	+0.00 .. +0.12	OK	OK
18	Strip read face to groove floor (rear-facing strip never rubs)	+0.250	+0.19 .. +0.31	+0.21 .. +0.29	+0.10 .. +0.40	OK	OK
19	Strip retention: crush-rib interference per side	-0.080	-0.13 .. -0.03	-0.12 .. -0.04	-0.15 .. -0.01	OK	OK
20	Read window vs optical axis (axial offset)	+0.000	-0.30 .. +0.30	-0.21 .. +0.21	-0.50 .. +0.50	OK	OK
21	Detent engagement (bump past the groove wall)	+0.350	+0.25 .. +0.45	+0.28 .. +0.42	+0.15 .. +0.55	OK	OK
22	Light-seal plug radial gap in the mouth funnel	+0.300	+0.05 .. +0.55	+0.13 .. +0.47	+0.00 .. +0.60	OK	OK
23	Handle face to housing skin (axial)	+0.500	+0.10 .. +0.90	+0.26 .. +0.74	+0.00 .. +1.00	OK	OK

CONTRIBUTORS

1. front boss end above P/L +0.10 ±0.05; rear boss top at P/L +0.00 ±0.05
Gap must stay ≥ 0 so the perimeter lap closes first (no visible P/L gap).
2. rear standoff seat below P/L +3.20 ±0.05; PCB thickness -1.60 ±0.16; front top boss end below P/L -1.30 ±0.05
Boss must never crush the PCB; PCB taken at IPC +/10 % here.
3. front inner face above P/L +5.80 ±0.05; rear standoff seat below P/L +3.20 ±0.05; PCB thickness -1.60 ±0.10; tact switch height -5.00 ±0.10; cap: stem tip to flange top -1.
Play $> 0 \Rightarrow$ the housing can never pre-press the switch.
4. cap: flange top to cap top +3.05 ±0.05; flange-top position (= face - play) -2.45 ±0.40
Stays proud of the overlay for a tactile press.
5. lens pocket depth +0.95 ±0.05; lens thickness (PMMA) -0.80 ±0.05; lens tape (3M 9471LE 0.05) -0.05 ±0.01
Lens must not push the overlay up (max 0.02 proud accepted).
6. recess depth +0.25 ±0.03; overlay 0.175 PC + 0.05 adhesive -0.23 ±0.03
Overlay edge must not stand proud (lift / peel).
7. door recess depth +1.25 ±0.05; door plate thickness -1.20 ±0.05
0.05 proud to 0.15 sunk accepted.
8. recess width / 2 +18.65 ±0.05; door width / 2 -18.50 ±0.05
Visible gap; door must drop in freely.
9. bump reach from arm face +0.80 ±0.05; arm-to-edge clearance -0.30 ±0.05; door float in recess +0.00 ±0.15; opening edge position +0.00 ±0.05
Engagement > 0.2 ; max deflection sets the latch strain (see note).
10. step face offset (front) +1.00 ±0.03; tongue outer offset (rear) -1.10 ±0.03; front/rear profile mismatch +0.00 ±0.05
Tongue 0.90 0/-0.05 makes the real minimum 0.02 (+0.05 thinner tongue); zero-clearance local touch is acceptable.
11. step depth (front) +1.70 ±0.05; tongue height (rear) -1.50 ±0.05
Tongue must never bottom out.
12. half opening height +2.00 ±0.05; half receptacle height -1.63 ±0.03; receptacle centre vs P/L (seat, PCB, part) +0.00 ±0.20
Plug must enter without touching the housing.
13. half opening width +4.80 ±0.05; half receptacle width -4.47 ±0.03; PCB lateral float (crush-rib clearance + outline) +0.00 ±0.15; receptacle placement on PCB +0.00 ±0.10
14. (hole - screw) / 2 +0.35 ±0.05; PCB float +0.00 ±0.15; hole position on PCB +0.00 ±0.10; boss position +0.00 ±0.05
Screw must pass without loading the PCB sidewalls.
15. groove width / 2 +7.15 ±0.03; tray width / 2 -7.05 ±0.03
Tray slides freely but cannot wander: ± 0.15 max lateral float.
16. groove depth (floor to block top) +3.30 ±0.05; tray thickness -3.10 ±0.05
Tray cannot tilt more than -0.3 mm over its 34 mm length.
17. pocket width / 2 +3.05 ±0.03; strip width / 2 -3.00 ±0.03
Pocket is located by the pocket; confirm strip width tolerance (assumed ± 0.05).
18. pocket recess (tray underside to pocket ceiling) - strip +0.25 ±0.03; strip thickness +0.00 ±0.03
Strip is held up against the pocket ceiling by the crush ribs; 0.25 air gap to the block floor.
19. pocket half width - rib reach +2.92 ±0.03; strip half width -3.00 ±0.03
Negative = interference: the strip snaps in and cannot drop out of the inverted pocket.
20. tray front face on the end stop (detent preload) +0.00 ±0.00; end stop to aperture (same part, HS-105) +0.00 ±0.05; tray front face to pocket wall +0.00 ±0.05; strip: from Block float on its screws does NOT enter: stop and aperture are on the same part. Read pad must be $\geq 2.0 + 2 \times 0.3 = 2.6$ mm.
21. bump reach beyond tray side +0.45 ±0.05; tray lateral clearance -0.10 ±0.05
Click is felt; arm strain $1.5 \text{ t d} / L^2 = 0.4 \%$ at max (ABS limit $\sim 2.5 \%$).
22. plug gap (design) +0.30 ±0.05; funnel profile +0.00 ±0.05; tray lateral float +0.00 ±0.15
Plug never rubs the funnel; 2-turn labyrinth for ambient light.
23. design gap +0.50 ±0.00; optical block float on screws (Y) +0.00 ±0.20; block end stop to housing skin (housing) +0.00 ±0.10; tray: from

LEV RAM LIFESCIENCES

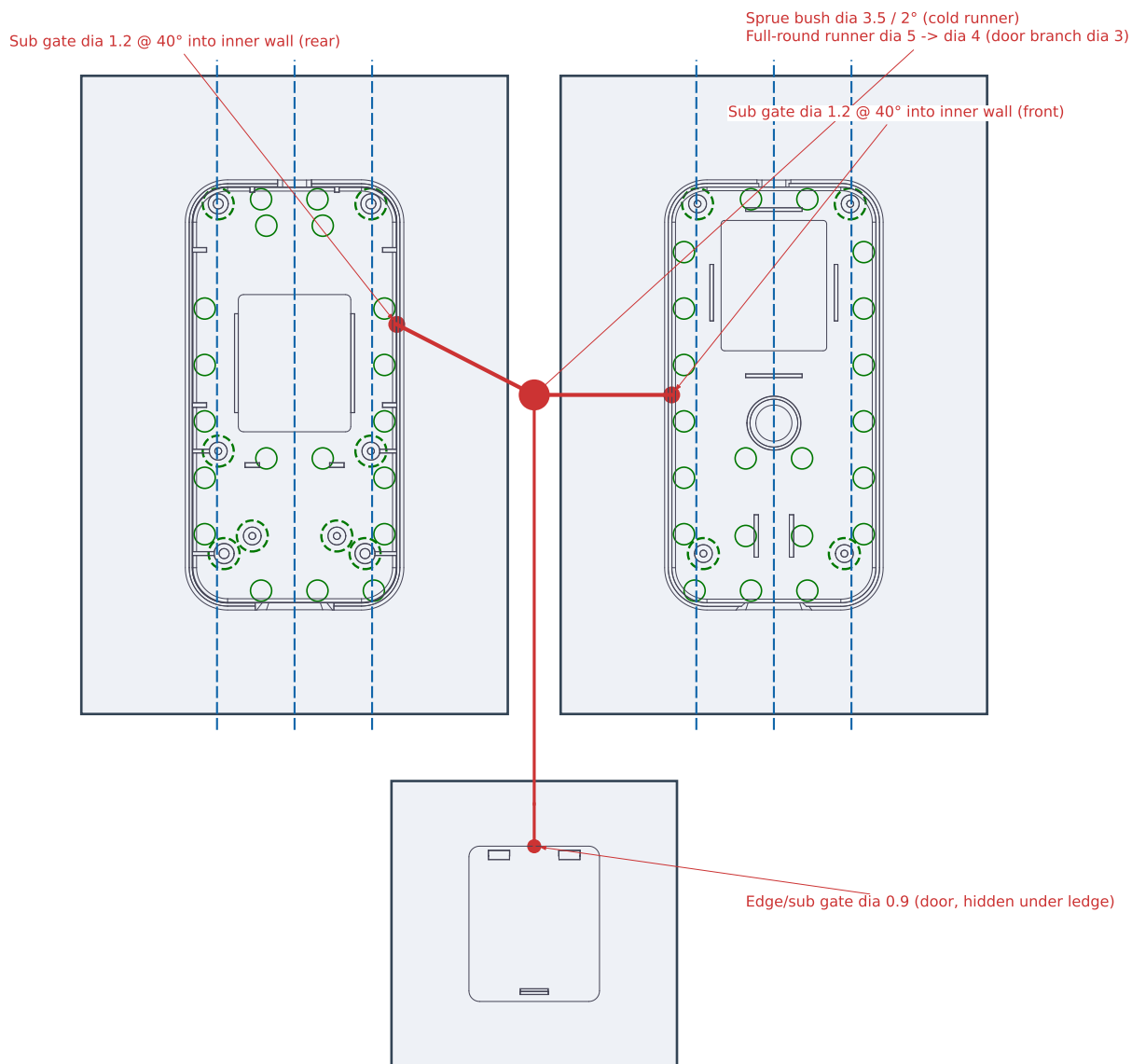
TOLERANCES, FITS & STACK-UP ANALYSIS

HemoSure Hb meter casing

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

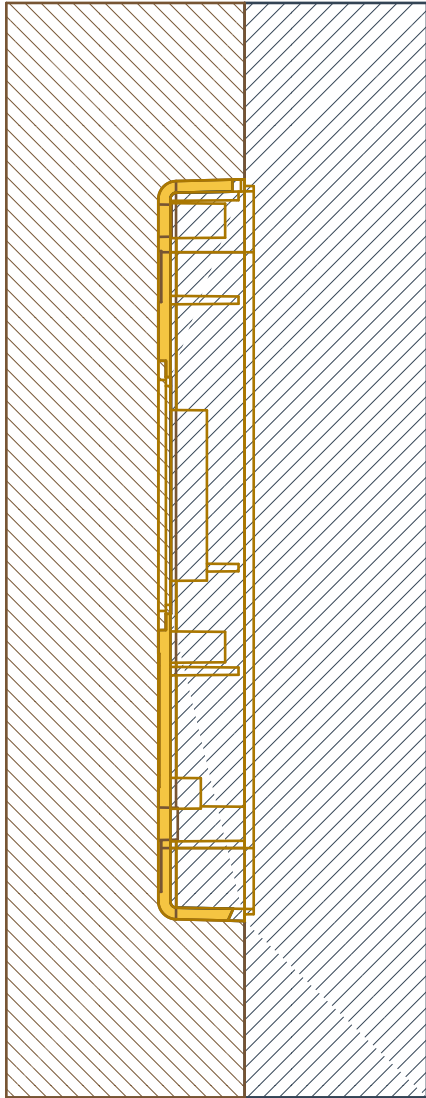
PART All moulded parts	DWG NO. HS-110	REV A
MATERIAL ABS	SCALE -	SHEET 7 / 11
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)
MASS -	DATE 2026-09-28	STATUS DRAFT - DFM review

Mould base 350 x 400 (LKM/Futaba FCI 3540 eq.) - moving half (core side) plan



REAR HOUSING - core (blue /) + cavity (brown \), part yellow

Section $x = 0$ at 0.8:1; mould opens along Z (horizontal on this sheet)

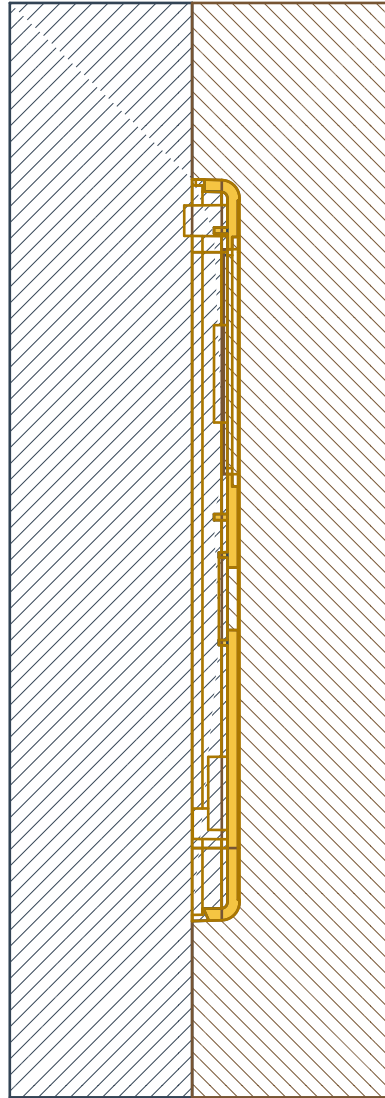


core bbox [120.6, 180.9, 42.4] cavity bbox [120.6, 180.9, 39.4] (mm)

rays 98882 | undercut rays 0 | out-of-order steel 0

FRONT HOUSING - core (blue /) + cavity (brown \), part yellow

Section $x = 0$ at 0.8:1; mould opens along Z (horizontal on this sheet)



core bbox [120.6, 180.9, 36.0] cavity bbox [120.6, 180.9, 33.0] (mm)

rays 100140 | undercut rays 0 | out-of-order steel 0

NOTES

1. Inserts are the full block 15 is the part (x 1.005). Toolmaker adds: pocket fits (H7/g6 to the plates), locks / taper interlocks (4 x 5" side locks per insert), ejector & sleeve holes, cooling drillings, vents, sprue/runner/gates.
2. Shut-offs: USB-C opening and cuvette mouth are formed by core steel shutting off on the P/L - give them 1-3" in the tool if the T1 shows scuffing (geometry change ≤ 0.03 , inside the opening tolerance).
3. Leave steel-safe on core: pilot pins -0.03, button hole -0.05, window/lens/overlay pockets -0.05, boss ends +0.05.
4. Polish core SPI B-2; cavity EDM texture VDI 24 after dimensional sign-off; lens pocket floor SPI A-3.

LEVRAM LIFESCIENCES

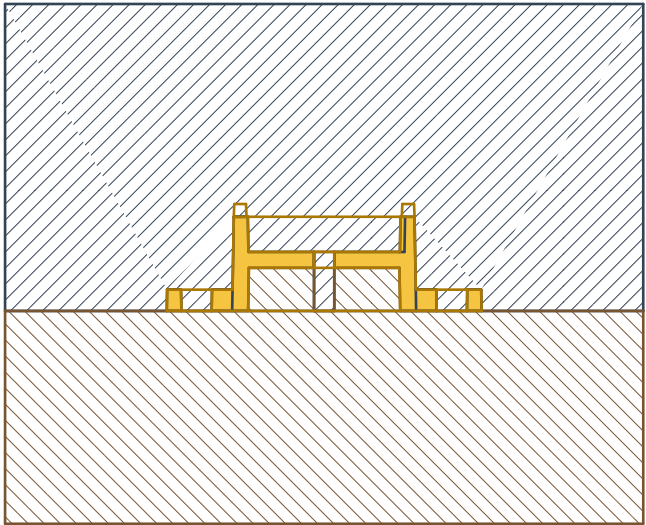
HemoSure Hb meter casing

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

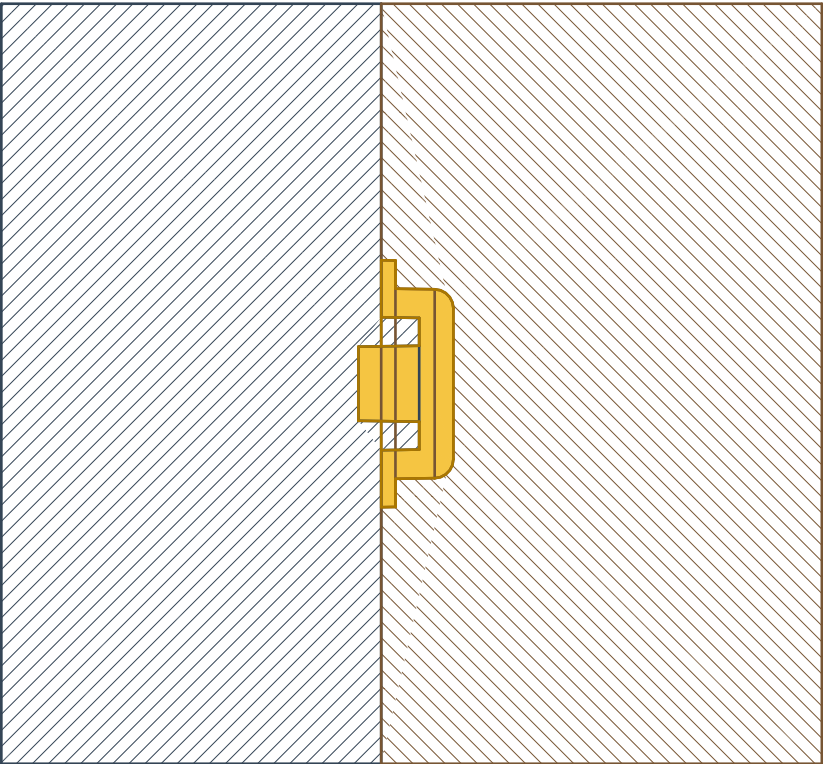
MOULD TOOL A: CORE / CAVITY SECTIONS

PART Tool A inserts	DWG NO. H5-302	REV A
MATERIAL 1.2311 / 1.2738	SCALE 1:1	SHEET 9 / 11
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)
MASS -	DATE 2026-09-28	STATUS DRAFT - DFM review

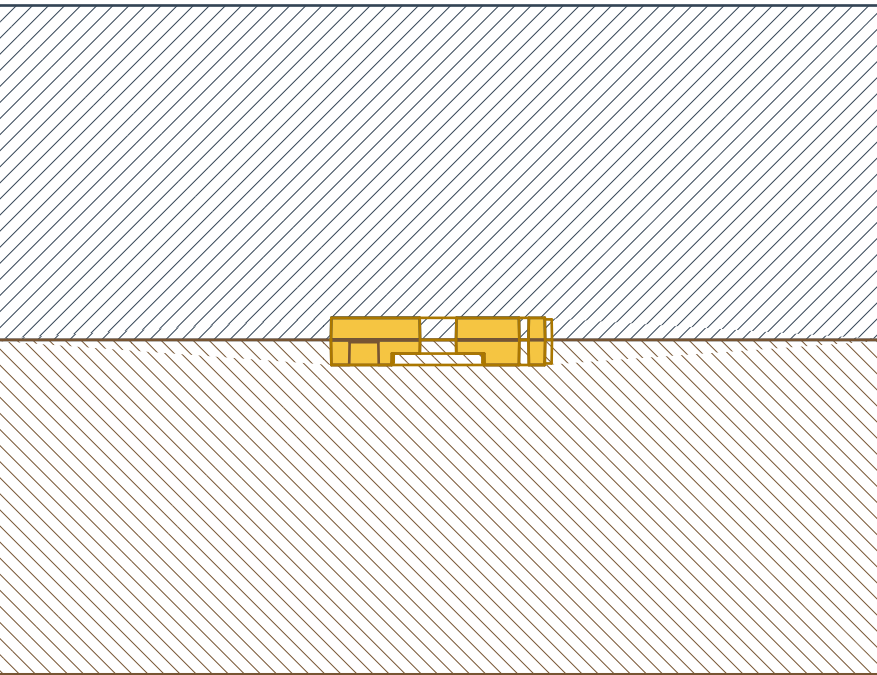
OPTICAL BLOCK - core / cavity (1.4:1)



BUTTON CAP - core / cavity (2.5:1)



STRIP TRAY - core / cavity (2.0:1)



rays 10809 | undercut 0 | out-of-order 0

rays 2117 | undercut 0 | out-of-order 0

NOTES

- 1. All three parts black -> one family tool; if the button must be another colour, split it to its own 1-cavity tool.
- 1a. Tray + block are a matched pair: T1 fit check = tray slides with <= 2 N, clicks home, no rattle > 0.15 mm.
- 2. Optical block groove is the optical-path datum: hold 14.30 +0.05/0 and floor 1.50 ±0.05.
- 3. Heat-stake pins dia 1.2: vent the pin tips (0.01) to avoid short shots.
- 4. Verify opacity on T1 parts: no visible light through 2 mm with a torch in a dark room; spectrometer < 0.1 %.

Item	Tool B data
Cavities	button + optical block + strip tray, black ABS
rays 11573 undercut 0 out-of-order 0	
Parting	Button: flange u/s. Block: u/s. Tray: P/L z 14.2
Shrink	0.50 % (in insert STEP)
Feed	Sub gate dia 0.8 (button, on stem side), sub gate dia 1.0 (block, on ear underside)
Ejection	Button: sleeve on stem; block: 4 pins on ears + 2 pins on the end walls
Polish	Groove, tray faces SPI B-1 (sliding)
Proj. area	~7.5 cm² -> 30-50 t press
Feed (tray)	tray: sub gate dia 1.0 on the handle underside
Shot	5.5 g parts + ~3 g runner

LEVRAM LIFESCIENCES

HemoSure Hb meter casing

MOULD TOOL B (1+1+1, BLACK ABS)

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART	DWG NO.	REV
Tool B: button + optical block + tray	HS-303	A
MATERIAL	SCALE	SHEET
Inserts 1.2311 P20 / cavity 1.2083 for polish	2:1	10 / 11
FINISH	UNITS	PROJ.
-	mm	1st angle (ISO E)
MASS	DATE	STATUS
-	2026-09-28	DRAFT - DFM review

A

B

C

D

E

F

AUTOMATED CAD CHECKS (this revision)

Part	Vol. cm³	Mass g	Rays cast	Undercut rays	Steel out-of-order	Faces < 0.2° draft (>1 mm)	Parting z
Rear Housing	21.25	22.3	98882	0	0	0	14.2
Front Housing	14.28	15.0	100140	0	0	0	14.2
Battery Door	1.97	2.1	26150	0	0	0	1.2
Button Cap	0.29	0.3	2117	0	0	0	18.8
Optical Block	2.01	2.1	10809	0	0	0	7.0
Strip Tray	2.95	3.1	11573	0	0	0	14.2

Assembly interference check, all part pairs incl. PCB/TFT/USB-C/tact/LiPo/strip/tray/LED board: [['optical','tray', 0.022], ['tray','REF_hb_strip_6x30x0.5', 0.164]] (optical x tray = detent preload; tray x strip = 0.08 crush-rib retention, both by design)

METHOD

- How the checks work (hemosure_casing.py + mold_tools.py, re-run on every change):
- Undercut: vertical rays on a 0.25 mm grid; any ray crossing the part in 2+ intervals = trapped steel (validated on a test part).
 - Inserts: rays through core, cavity and part; order bottom->top must be one steel half, part, other half (no interlock).
 - Draft: every face's angle to the draw axis; faces taller than 1 mm with < 0.2° are listed (target: none).
 - Stack-ups: 23 worst-case + RSS stacks, sheet 7. All pass worst case at this revision.

DESIGN RULES APPLIED

Wall 2.00 nominal (ABS range 1.2-3.5); thinnest: lens-pocket ledge 0.80, door ledge 0.75, lap skirt 1.00.
Ribs 1.0-1.2 (50-60 % of wall), rib height <= 3x wall except PCB posts (gusseted, sleeve-ejected).
Boss OD 5.0-5.6 / hole 2.0-2.9: boss wall 1.35-1.5 (<= 0.75 x wall) -> no sink on the rear skin.
All inside corners R >= 0.5 at wall/floor (R1.0 modelled at the main wall-floor junction).
Flow length / wall (L/t) max ~42 from a single side gate - well inside ABS limit (~150).
Critical cosmetic: front overlay field, rear skin, P/L line. Non-cosmetic: interior, optical block.

MOULDING START POINT

Process window (start point): melt 230 °C, mould 55 °C, injection speed medium, hold 60-70 % of fill pressure for 4-5 s, cooling 9 s, ejection at part temp < 85 °C. Gate freeze study to set the hold time. Regrind <= 15 % on cosmetic parts.

DATA PACKAGE

Files (all regenerated from cad/hemosure_casing.py - change a parameter, re-run, everything updates):
cad/HemoSure_*.step (parts + assembly) mold/HemoSure*__{CORE,CAVITY}.step (steel size) drawings/*.pdf
cad/tolerance_stack.py (stacks) cad/dfm_checks.py (draft) mold/mold_check_report.json (ray undercut / insert order)

VALIDATION

- T0/T1 TRIAL CHECKLIST (for the tool room)
1. Dry ABS 80 °C / 3 h; dew point <= -20 °C.
 2. Short-shot study -> fill balance of the family tool.
 3. Measure the critical-dims table (sheet 6) on 5 shots/cavity;
 Cpk >= 1.33 on: boss pattern, PCB seat, window, lens pocket.
 4. Assemble with a real PCB + TFT + LiPo: USB-C plug insertion,
 button click (no pre-press, 0.25 travel), door open/close x 50.
 5. Drop test 1.0 m x 6 faces (IEC 61010-1 / 60068-2-31 style).
 6. Light-leak test of the optical cell with the strip tray inserted.
 7. Disinfectant wipe x 500 (70 % IPA, 0.5 % NaOCl): no ESC cracks
 at bosses (if cracking: switch to PC/ABS, shrink 0.5-0.7 %).
 8. Then texture the cavities (VDI 24) and re-check P/L step.

LEVRAM LIFESCIENCES

HemoSure Hb meter casing

DFM VERIFICATION & MOULDING SPECIFICATION

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART All moulded parts	DWG NO. HS-310	REV A
MATERIAL ABS	SCALE -	SHEET 11 / 11
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)
MASS -	DATE 2026-09-28	STATUS DRAFT - DFM review